

YUVARAJ S

B.Tech Artificial Intelligence and Data Science

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CAREER OBJECTIVE

Motivated B.Tech Artificial Intelligence and Data Science student with hands-on experience building and deploying AI-driven applications using LLM APIs, RAG pipelines, and prompt engineering. Proficient in Python, with practical exposure to vector databases, REST API integration, and AI-assisted development workflows. Eager to contribute as an Associate AI Engineer, building production-ready AI features and rapidly prototyping proofs of concept.

EDUCATION

Higher Secondary Education

2022-2023 | Score:69%

Government Higher Secondary School, Vathalmalai, Dharmapuri

Bachelor of Technology

2023-2027 | CGPA: 6.52

Bannari Amman Institute of Technology, Sathyamangalam, Erode

TECHNICAL SKILLS:

- **Programming Languages:** Python, Java, C
- **AI / LLM:** LLM APIs (OpenAI, NVIDIA NIM), Prompt Engineering, RAG Pipelines, Multimodal Retrieval, NLP, Computer Vision
- **Frameworks & Tools:** FastAPI, Flask, CrewAI, REST API Development, Git & GitHub
- **Databases:** MySQL, MongoDB, Vector Similarity Search
- **Web Technologies:** HTML5, CSS3, JavaScript
- **Core CS:** Data Structures and Algorithms, OOP, DBMS

PROJECT EXPERIENCE:

Voice-Based AI Study Assistant

- Developed a voice-enabled AI study assistant capable of analyzing spoken input and extracting study topics automatically.
- Implemented interactive quiz and assessment features to improve learning engagement.
- Enabled real-time audio communication using speech processing techniques.

Visual Search Engine for Jewelry using Multimodal RAG

- Built a Retrieval-Augmented Generation pipeline integrating LLM APIs to generate product descriptions from retrieved jewelry data
- Designed vector similarity search across image, sketch, and text inputs for accurate multimodal retrieval
- Structured prompts and parsed LLM outputs to produce consistent, structured product descriptions

PlacementPal — AI-Powered Placement Preparation Platform

- Built a multi-agent AI platform using LangGraph, CrewAI, and LangChain to generate personalized study plans and company-specific interview intelligence
- Implemented a RAG pipeline over ChromaDB for semantic document retrieval, with a FastAPI backend and async MongoDB persistence
- Integrated the Tavily Search API for real-time company hiring insights within a two-phase stateful LangGraph pipeline

CropCopilot – AI-Powered Agricultural Intelligence System

- Integrated NVIDIA NIM LLM APIs with CrewAI agents to combine RAG-based knowledge retrieval with Text-to-SQL querying
 - Built and deployed a FastAPI backend serving real-time AI-generated insights, including error handling for API/model edge cases
 - Rapidly prototyped a working proof of concept combining structured data queries with unstructured knowledge retrieval
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CERTIFICATIONS

Prompt Engineering — Navigate Labs

- Completed a 4-week intensive hands-on training covering prompt design techniques, few-shot and chain-of-thought prompting, and structuring inputs for reliable LLM outputs.

Retrieval-Augmented Generation (RAG) — Navigate Labs

- Completed a 4-week intensive hands-on training covering RAG pipeline design, vector embeddings, and semantic retrieval for grounding LLM responses in external knowledge sources.

Voice Artificial Intelligence — Navigate Labs

- Completed a 4-week intensive hands-on training covering speech-to-text and text-to-speech pipelines, real-time audio processing, and building voice-driven conversational AI systems